

Challenges in Serious Game Design and Development: Educators' Experiences

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Abstract

Background. Relatively little is known about the role of **educators in serious game design and development** and their experiences with **serious game implementation**. We investigate educators' perceived **challenges** deriving from their involvement as subject matter experts during the serious game development trajectory.

Methods. A secondary analysis of data collected through an exploratory survey about serious game design and development approaches was carried out. The sample included 41 educators from post-secondary education institutions across North America. An in-depth analysis of qualitative data revealed educators' roles in game development, the challenges they faced, and the **strategies** they deployed in serious game design, development, and implementation.

Results. Educators, as serious game designers, perceived challenges to be administrative, design-related, attitudinal, and communicative. Strategies deployed to overcome challenges during the concept development, pre-production, and production stages of game design include the creation of games that balance learning and fun, and enhanced team collaboration through cultural mediation. During the post-production stage, although challenges are acknowledged and some strategies, such as improving the usability of a game, are deployed, a clear pattern in challenges and mitigating strategies could not be observed.

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Conclusion. Serious game design and development can be improved by nurturing diversity of ideas and adopting creative design and development methodologies. Serious game implementation can be improved by devising effective administrative and attitudinal strategies, and incorporating diversity of ideas into target curricula. Additionally, clear directives about usability should be devised, and academic objectivity towards serious games needs to be created. Strategies to achieve these goals should focus on developing trust between target users, the technical development team, and educators as serious game implementers.

Keywords

active learning, educational technology, experiential learning, game-based learning, game design process, game development, post-secondary education, serious games, simulations

Introduction

The idea of purposing play for learning can be traced back to Plato's works (Wilkinson, 2016). Yet today's powerful computing infrastructure opens up new opportunities for game-based learning. Educators are using game-based learning as a teaching approach in K-12 and post-secondary education because of its potential to aid teaching a variety of subjects in an entertaining and engaging way (Annetta et al., 2009; De Gloria et al., 2014; Iten & Petko, 2016; Muntean, 2011), and to promote learner independence and empowerment (Bleumers et al., 2013; de Freitas, 2006; Stewart et al., 2013). This, coupled with a pedagogical shift from passive to active, experiential learning have increasingly led educators to explore innovative digital educational technology tools in post-secondary education. Such tools include serious games (SG).

SG are games that promote learning and behavior change through entertainment, and are used in various sectors such as education, health, business, along with the armed forces and the aviation industry (Brandão et al., 2012). In their typology of simulation games, Lean et al. (2006) distinguished between non-computer and computer-based games. Role play is an example of a non-computer-based game whereas flight simulations would be an example of computer-based games. In this study, we focus on computer-based games, and define SG as digital games designed purposefully to link game objectives with learning objectives, and to teach players content or skills in an engaging and entertaining way.

A well-designed SG provides a medium in which learners apply subject matter to real-world contexts (Guillén-Nieto & Aleson-Carbonell, 2012). Playing SG, learners typically develop a deep interest in the educational content of the game, which leads to improved learning outcomes (Wronowski et al., 2020). Further, SG implementation in classroom and online courses has been shown to improve retention rates and encourage creative exploration (Bergeron, 2008) – outcomes of significant interest to academic administrators and educators alike (Clarke et al., 2013). Despite these positive

outcomes, little attention has been paid to the multidisciplinary nature of Serious Game Design and Development (SGDD) and participatory SGDD design processes (Korhonen et al., 2017), especially to the roles and associated tasks of SGDD multidisciplinary team members. This is the gap we aim to address through this research.

Aim and Research Questions

Despite the promise of SG in post-secondary education, there are challenges in developing highly effective SG. This can, to an extent, be attributed to insufficient understanding of the transitioning and multifaceted role of the educator, or subject matter expert (SME). The effectiveness of SG is limited when the educator is not thoroughly integrated into the design and development process; yet, the role of the educator, as arguably the most critical piece in learning, is an under-studied subsection of game-based learning research. Hence, we argue that much more focus must be placed on the role of the educator if SG are to be widely adopted and become mainstream. In this study, we explore educators' perceived challenges and strategies deployed in the end-to-end development cycle of SGDD, and address the following two research questions:

1. What types of challenges or barriers do educators experience when designing, developing, and implementing serious games at post-secondary education institutions?
2. What strategies do educators employ, if any, when designing, developing, and implementing serious games at post-secondary education institutions?

The rest of this article is organized as follows: we review the literature and introduce our conceptual framework and methodology. Then we present our findings and discuss them. Finally, we discuss the potential limitations of this study, and suggest directions for future research.

Literature Review

Philosophical approaches to SGDD have been likened to art and science. Easterly (1978) proposed a distinction between model building and model discovery approaches, where the former is in line with traditional instructional design practices, and the latter is an artistic endeavor not restricted by the rigidity of the model-building approach. Hall (2014) traced the aspects of reality, functionality and emotional engagement in business simulations to fine art movements, and parallelized the progression from business modeling simulations to abstract models focusing on learning and engagement, to the movement from realism to surrealism. Klabbers (2006) suggested that approaches in gaming and simulation represent both design and analytical sciences.

The need for engaging learning games has led to the development of design frameworks and processes, rooted in traditional instructional design models. These can be

categorized into frameworks and processes focusing on the balance of elements such as the *Triadic Framework* (Rooney, 2012), the *Balanced Design Framework* (Groff et al., 2015), the *Game Rules scenario Model* (Zarraonandia et al., 2015), or the *Game-based Learning Framework* (Staalduinen & de Freitas, 2011), or concentrating primarily on iteration, for example the *9 step process to game design* (Duke, 1980), the *Transdisciplinary Model*, (Arnab & Clarke, 2017; Arnab et al., 2015), the *Tandem Transformational Game Design* (To et al., 2016) or the *iMPOS²inG Model* (Smith et al., 2018).

While these design approaches depict abstract representations of proposed SGDD strategies without much practical, step-by-step guidance for multidisciplinary teams, none focus their attention on the role of the educator participating in SGDD. As a relatively new discipline, the study of SG is still refining itself with the development of theoretical, conceptual, and methodological approaches along with considerations of the broader purpose of their existence. The theoretical development of a comprehensive approach to SGDD is fragmented (Mayer, 2012), and similar approaches adopted across sectors are scattered across academic journals and conference proceedings.

Kirkley et al. (2007) called for new types of game design strategies that will allow the transformation of learning goals and evaluative markers into serious, entertaining gameplay, in which instructional designers will play a pivotal role in translating game goals into instructional goals, and develop models that will link aspects of art and science into SGDD. However, they did not to acknowledge or address educators' importance and contribution in the SGDD process as project initiators, subject matter experts, evaluators, and implementers of SG, consequently reinforcing the replication of current SGDD processes. Further, they perceived subject matter expertise as something static that is easily transformed into learning goals and incorporated into a game. Hence, they failed to account for the contribution of educators to SGDD, consequently providing no practical guidance to individual roles of multidisciplinary teams.

The Role of the Educator in the Serious Game Design and Development Process

A deeper understanding of the educator role is important for SG development as this will enable researchers to refine existing design approaches and give rise to new frameworks and processes potentially leading to more widespread, successful SG development and implementation in education.

SGDD is a highly complex process that requires collaboration and input from multidisciplinary team members (Aleem et al., 2016) including subject matter experts, instructional designers, game designers, developers, project managers, graphic designers, and sometimes actors. Specifically, SG frameworks and methodologies in post-secondary education follow, to a large extent, a user-centered design approach (Rankin et al., 2008). As subject matter experts, educators ensure that content is accurate and that the SG is pedagogically sound and suitable for use in education (Pitts & Ashwin, 2005). They are involved in the user-centered design phases due to their interaction with the intended target audience. In fact, educators are the only SGDD team members who interact with the target audience.

Through our review of the literature, we found very little empirical research to date that examines the role of subject matter experts in SGDD. For example, Zook and Riedl (2013) did not even distinguish between roles in their wide scale analysis of development process trends in time-constrained game design. Similarly, Ruggiero and Watson (2014) examined the game development process in business, government, and academia, focusing on the game design experience using reflective practice. However, they did not differentiate between roles or describe educators' perspectives in their data and analysis. Aleem et al. (2016) acknowledged the difficulties associated with the multidisciplinary nature of the SGDD process, but did not distinguish between team roles, such as the role of subject matter experts in serious game design.

The Role of the Educator in Serious Game Implementation

Another research gap is how educators, as part of the SGDD design team, approach SG implementation, which refers to the wider use of SG in post-secondary education institutions. Most existing studies liken SG implementation with adoption, failing to examine implementation as part of the SGDD process during post-production. Similarly, most studies on SG adoption focused on K12 education (De Grove et al., 2012; Huizenga et al., 2017; Kebritchi, 2010; Kenny & Gunter, 2011; Proctor & Marks, 2013; Ucus, 2015), but post-secondary education evidence is limited with existing research being on a small scale.

Lean et al. (2006) and Moizer et al. (2009) examined academics' use of simulation-based teaching approaches and their perceptions of adoption barriers. They identified that the use of simulations was influenced by attitudes towards suitability and risk for learning. In their subsequent investigation examining the interrelationship between barriers and impact of other factors, they identified time and resource availability, training and development, informal learning support, and access to networks and secondary information sources as mechanisms to overcome adoption barriers.

In their study investigating students' and educators' attitudes towards SG implementation as a teaching and learning tool, Kapralos et al. (2011) found that unavailability of appropriate SG was a common reason for educators not using SG. Games that were selected were those where SGDD closely aligned with course outcomes. They also reported that SG implementation was limited because educators, already pressed for time, were reluctant to put in the extra effort needed to integrate SG into the curriculum. Similarly, Noraddin et al. (2013) investigated the perceptions of educators towards SG implementation, and confirmed that educators had favorable attitudes towards SG implementation. While gender, age, discipline, and years of teaching did not influence attitudes, educators who played digital games themselves had a more favorable view of SG implementation.

Other challenges in implementing SG in post-secondary education, identified by Chan and Zary (2019), were limited time for teaching skills development, limited resources to deploy new teaching methods, and the lack of knowledge about SG as teaching tools. Swier and Peterson (2018) examined implementation of 3D SG in English language teaching in Japan and identified the availability and usability of

computer resources as the main challenge. They also found that participants who implemented SG did so in secret because their peers' perceived SG as having limited academic value. Lack of administrative support was also identified as a barrier.

In summary, there is limited understanding of the educator role as a member of a multidisciplinary team involved in the SGDD process. What is more, evidence regarding the challenges educators face in SGDD, including implementation, is limited. Current research highlights that educators face institutional (e.g. lack of administrative support, time and resources), social (e.g. peer pressure), and personal barriers (e.g. lack of knowledge or professional development opportunities) where there is a research gap regarding their role in developing SG, their experiences as game designers and collaborators on SG multidisciplinary design teams and in embedding SG into the curriculum. We address this research gap through an in-depth exploration of educators' perceptions of their involvement in SGDD, including the stage of implementation.

Conceptual Framework

In our work as SGDD practitioners at a large, urban, higher education institution, we are developing immersive educational games and simulations offered freely as open educational resources (OER) that have been played more than 180,000 times in over 25 counties. Our design approach strives to align game theory and gameplay with instructional design, while our design team is encouraged to adopt a creative methodology that enhances communication between team members.

Multidisciplinary team members' task allocation of our SGDD approach is illustrated in Table 1. We have adopted Hirumi and Stapleton's (2008) stages of SGDD, consisting of *concept development* involving pitch, *pre-production* involving game design creation and production planning, *prototype and production* where the concept is formed into a product, and *post-production* focusing on game evaluation, refinement, and implementation. We have added the pre-stage of *preparation* due to the nature of fundraising for the creation of OER materials, following the design science research methodology paradigm suggested by Peffers et al. (2007).

As can be seen in the table, the educator is the only team member involved in all stages of SGDD. In the preparation stage, the educator occasionally collaborates with the instructional designer, depending on availability and resources. In the stages of concept development, pre-production and production, the educator must find effective ways to communicate, often highly complex content, to the multidisciplinary team, ensuring that learning and game objectives are effectively integrated into a game. Most interestingly, the multidisciplinary team is not involved in the implementation stage, and the educator is tasked with finding effective strategies to overcome implementation challenges through effective strategies for SG promotion, adoption, and curriculum integration.

Our experience has further informed us of the complexities of SGDD and made us very aware of a major barrier to in-house SG design: lack of resources. Additionally, in-house development presents several challenges such as putting new demands on faculty, recruiting gaming specialists, and the resulting financial costs. We have

Table 1. Roles of Multidisciplinary Team in Serious Game Design and Development.

Game development stages					
	Preparation	Concept development	Pre-production	Production	Post-production (implementation)
Educator	Identification of learning needs and objectives Proposal Writing Project Budget	Drafting learning objectives Specifying learning outcomes	Assessing linking of Learning Objectives with Game Objectives Providing content Further developing type of skills/ behavior to be demonstrated/ practiced through the game	Assessing linking of Learning Objectives with Game Objectives Assessing alignment of game content and flow in prototype/ script etc. Piloting with a sample of target audience and assessing if learning objectives are being met Usability testing	Integrating it into teaching/ training Evaluation Use/Adoption
	Brainstorming learning needs with educators Sometimes involved in proposal writing	Identifying learning objectives for the game	Assessing linking of Learning Objectives with Game Objectives Providing feedback on content	Assessing alignment of Game Design Elements and Objectives with learning outcomes	No input
Game Developer	No input	Advises on game mechanics and technology	Linking Learning Objectives & Game Objectives into game mechanics. Prototype design	Developing game/ incorporating revisions	No input
others*	No input	Minimal input	Writing script Designing background or film setting Working on the script	Final assessment/ revisions to script/ background/ filming etc.	No input

*The role of **others** involved in SGDD depends on the game type and budget and may involve a scriptwriter, editor, script supervisor, actors, graphic designer, etc.

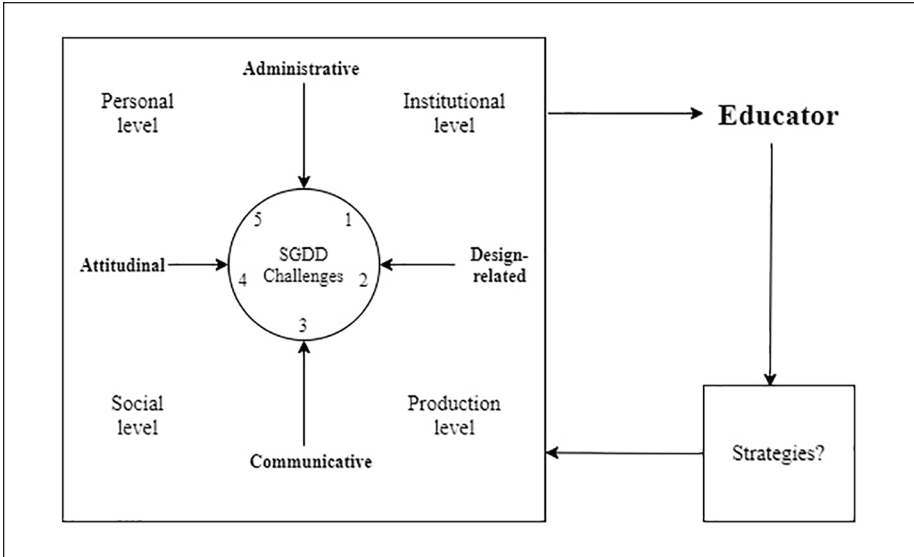


Figure 1. Challenges transformed into strategies by educators during the 5 stages of SGDD.

observed that limited selection, high prices of SG, and requirements of compulsory purchase of textbooks by publishing houses have led some faculty to design their own games. These educators are involved in, and contribute to, all stages of serious game design and development (SGDD) and are active in implementing these new games in post-secondary education.

Based on the review of the literature, we postulate that the challenges educators face are multifaceted and interrelated. These are barriers deriving from administrative challenges at the institutional level, communicative barriers at the social level, and attitudinal barriers at the personal level (see Figure 1). Similarly, based on our experience, we identify an additional barrier that poses design-related challenges at the production level.

Through our empirical study, we attempt to gain detailed understanding of the pieces in this conceptual framework by exploring educators' perceptions of the challenges experienced and the strategies deployed, when participating in SGDD as subject matter experts within a multidisciplinary team.

Methodology

This study was based on a secondary analysis of an exploratory survey about SGDD approaches and processes, which was part of a wider project aiming to develop a practical SGDD methodology and design guide (Digital Education Strategies, 2018), situated within a design science research methodology framework. Specifically, a supplementary, in-depth analysis (Heaton, 2004) of qualitative responses from a sample of 41 educators in post-secondary education across North American institutions

Table 2. Characteristics of Participants.

Characteristics	No (%) of respondents
<i>Game development experience (N=41)</i>	
6 or more games	16 (39%)
3-5 games	12 (29.3%)
Novice	13 (31.7%)
<i>Taught subjects (N=37)</i>	
Computer science	21 (56.8%)
Science/ health science	7 (18.9%)
Social science/ education	4 (10.8%)
Business & management	3 (8.1%)
Visual arts	2 (5.4%)
<i>Teaching approaches (N=35)</i>	
Combination of approaches	13 (37.1%)
Experiential/ project based	10 (28.6%)
Interactive/ inductive	4 (11.4%)
Lecture	4 (11.4%)
Other	4 (11.4%)

was isolated from the wider sample, because the unique role of educators in all stages of the SGDD process was an emerging issue beyond the scope of the primary study.

A snowball sampling approach was used in the main study. Using Opinio Survey Software, an invitation for participation with a link to the online survey was sent out to major organizations involved in the development of digital learning including: The Canadian Association for University Continuing Education, The Ontario Council for University Lifelong Learning, The Ontario Universities' Council on e-Learning and the Educational Developers Caucus across North America. In turn, these organizations distributed the survey to their members. This study was approved by our institution's research ethics board, and participants provided their consent electronically before completing the survey.

Descriptive statistics were calculated using SPSS v24 for quantitative items. The responses to the open-ended items were imported into NVIVO. Our conceptual framework was used as an analytical lens during the coding of data and aided the identification of subthemes. Specifically, data were coded by challenges identified in our conceptual framework and then recoded, identifying strategies deployed, and other emerging themes. The unit of analysis was educators' perceptions of challenges encountered during SGDD and strategies deployed to overcome them.

Forty-one educators responded to the survey. The majority (see Table 2) reported having been involved in the development of six games, almost one third had developed three to five games, and about one third were new to game development. Over half of the respondents taught computer science, over one quarter taught science or health-related science, one out of ten taught social science or education, three taught business or management studies, and two taught visual arts. Regarding teaching

Table 3. Frequency of Challenges as Perceived by Educators.

Challenges	Frequency	(%)
Institutional level: administrative challenges	13	36.1
Production level: design related challenges	9	25
Personal level: attitudinal challenges	9	25
Social level: communicative challenges	2	5.6
Other	3	8.3
Total	36	100

approaches, over one third reported to use a combination of approaches, followed by over one quarter that reported the use of an experiential or project-based approach, and over one in ten the use of either an interactive, lecture or other approaches.

Results

The findings highlight that educators involved in SGDD perceived challenges to SGDD and implementation to be administrative (13), followed by design-related (9), attitudinal (9), communicative (2), and other (3) such as gender dynamics, involving all stakeholders, and message development. These challenges fall within the institutional, production, social and personal levels respectively (see Table 3), and are, to a large extent, interrelated.

Challenges in SGDD

Administrative Challenges

Educators tended to perceive SGDD as an extremely time-consuming and demanding activity when it is added to an already heavy workload. The most reported challenge was “*getting resources (such as time or money) to develop the games*”. Participants acknowledged that game design is labor-intensive and involves “*funding, subject matter expert time and administrative hassles*”. The shortage of appropriate games and resources meant that educators had to expand their teaching and subject matter expert role to designer and developer, which was overwhelming for many. One participant pointed out: “*I would rather be the content expert and work within a team of designers*”.

A few also pointed to the lack of institutional support for educators’ training opportunities in using new technologies, and access to technology: “*I teach in a very rural state and sometimes bandwidth is a problem*”.

Regarding SGDD implementation, educators described challenges related to the broader context of teaching and learning in post-secondary education. Specifically, reference was made to the difficulties associated with the need for strong data to support success. Participants referred to the expectation of “*showing evidence of learning to appease stakeholders*”, as the game’s effectiveness must be demonstrated right from the start.

Design Related Challenges

Linking outcomes, game design, and evaluation was described as a very complex undertaking that posed design-related challenges. One participant noted: “*creating the dimensions and metrics for evaluation seems to be more challenging compared to other applications*”. Another commented “*establishing the right teaching goals and determining what is best done with a game, and what is best done with other educational means*” were the important issues when using serious games.

Few participants made reference to using methodologies from other fields such as agile methodologies, and the affinity diagram approach and iterative design methods: “*[what] worked: progressively enhance game environment features. Start small and build features using feedback*”.

It was noted that finding “capable game designers with all the requisite skills needed such as programming, graphics, collaborative problem solving, and narrative expertise” was very difficult. Another participant’s comment captures the skills and expertise required for SGDD and the multidisciplinary team’s demands:

Developing a complex learning game requires a lot of skill sets: you need to have a deep understanding of the conceptual material you are trying to teach, you need to have [an] advanced understanding of instructional design and learning theory, and in the case of an online game, you need to have the technical skills to actually build the game. Few people will have all of these skill sets. As such, working on a team becomes necessary. Putting together the right team requires financial resources and time and energy. The average instructor is not going to have access to these resources.

Very few educators had all the components required for the development of quality SG, namely, an understanding of SGDD, subject matter and curriculum expertise, a background in developing teaching and learning activities, and budget skills.

A further design-related challenge identified was that if SG are going to be successful learning tools, they need to balance learning and fun. The balance between learning and fun can be achieved by “*connecting the learning goals and skills with the mechanics in a non-superficial way*”, and by “*recognizing the minimum requirements to maintain user engagement with a given outcome/objective*”. Blending those key game elements with required content requires creativity. The games are also seen in need of being interactive in order to be engaging and provide “*individualized feedback for learners*”. As one participant explained:

I would say that the greatest challenge was balancing the pedagogical elements of the game with the elements that were intended to engage the students with the sometimes complex content in the game [. . .]. The element of the game that was intended to engage students was the solving of a mystery [. . .] that would bring players in contact with non-player characters [. . .], and with content related to the game’s primary mission gathering and synthesizing information about various impacts of climate change on the local watershed and how those factors [. . .] could impact the health of natural systems, the town’s future and individual human beings [. . .].

Achieving a balance between fun and learning in SGDD was regarded as a major goal by most participants, and described as *“balancing the natural desire to make it ‘fun’ with the need to cover the learning objective or subject”*. As one participant explained: *“Too many people try to make it too learning-heavy. The best learning games I’ve found were 10-20% learning and 80-90% fun”*.

Interestingly, some educators expressed their skepticism regarding the ability of SG to promote learning in their field. For example, games were perceived as limiting for some science programs that focused on certain topics. According to one science educator: *“My courses revolve around fluid integration of interactive techniques. As closed worlds, games are limited in their use. More to the point, they are more useful as homework options [. . .]”*. In this case, faculty perceived SG as confining learners to a virtual world, instead of immersing them in the field.

Further design-related challenges included considerations about the best way to present content in the game, tailoring the different game components for the target audience, tackling all age groups, maintaining engagement, and playtesting: *“you can playtest only once with the learners before they know the concepts”*.

Regarding SG implementation, the suitability and sustainability of an SG, as well as assessment evidence, were perceived as challenges to teachers wishing to integrate SG into their curricula.

Communicative Challenges

Working with others on a multidisciplinary team that requires cross-cultural collaboration was perceived as complex and demanding. Teamwork meant assembling the right team and making it function well. In turn, this meant finding and putting creative people on the team, getting their commitment to the project, and promoting communication between team members. Participants referred to the richness of multiple communication channels when working in teams. However, many reported that navigating those channels was time-consuming and difficult at times. Team collaboration was identified as an essential element of SGDD strategy with reference to the multidisciplinary nature of SGDD teams. Participants expressed the view that an effective team was a *“good and willing team”* with creativity and diverse expertise. According to one participant’s experience:

There was a challenge of bringing together different professions to create the virtual gaming simulation. We came with different ideas and perspectives. It was also our strength, without all the different team members we could not have created such an effective learning tool.

With regard to implementation, most of the participants believed that a great communicative challenge was to convince administration, management, and students of the value of SG as a learning tool due to their possible *“resistance to the idea”*, which led many to think of ways to *“figuring out the right way to implement the game and defining the scope”*.

Attitudinal Challenges

Using serious games as a teaching tool was perceived to be effective by many participants; nevertheless, because of implementation challenges, game-based learning as a teaching approach was often questioned by administrators and other educators. Participants noted that this put a demand on them for *“tracking and evaluating learning criteria, correct measurement of outcomes and connecting game play to focus on game objectives”*.

“Overcoming the traditional academic attitude towards fun”, and acceptance of SG by the wider academic community as practical and effective pedagogical tools that engage learners while delivering specific knowledge within the broader context of experiential learning was an attitudinal challenge of potential gatekeepers, especially during SG implementation. Most respondents reported a perceived bias of their colleagues, i.e. *“people who don’t play games”* who labeled game-based learning *“as not being serious”*. References were made to difficulties in *“selling the concept to administration”* or *“convincing students and other faculty”* of the advantages of implementing SG as learning tools.

While SG were seen as valuable supplementary learning tools by many, for example in programs such as nursing, where SG help players develop real-world clinical skills, participants emphasized that there had to be a strong fit between the skills developed in a game with a particular program of study. Faculty must see that fit clearly; otherwise, a game may be perceived as not *“serious”*, and therefore be counteractive to learning. One participant noted that this means games need to be *“tailored”* to account for target audiences. Another participant said: *“the barrier is the fact that it’s not mainstream yet and there is an inert hesitation or misleading conception that people have with “gaming” that hinders the ability to maximize the benefits from games in a learning setting”*. Hence, if SG curriculum implementation is to become mainstream, a shift from traditional academic attitudes is needed.

Effective SGDD Strategies

A critical point in SGDD is that the multidisciplinary design team has to find and adopt strategies overcoming administrative, design-related, communicative and attitudinal challenges if a SG is to be successfully designed and implemented throughout the whole SGDD process (see Figure 2).

Specifically, during the conceptualization of an SG idea the team has to be clear regarding the course or learning module objectives and match these with the game objectives in order to achieve a balance between learning and fun. The setting of learning objectives was regarded as essential to the need to design *“fun”* games, and experiential learning was the motivator for the use of SG. Educators expressed the need to provide learners with a simulation of reality within a SG, where one can apply acquired knowledge. Thus, setting appropriate learning objectives was seen as aiding the *“1. triggering [of] different cognitive processes that are not fully utilized using conventional methods. 2. Creating more engagement. 3. Add[ing]some game mechanics and*

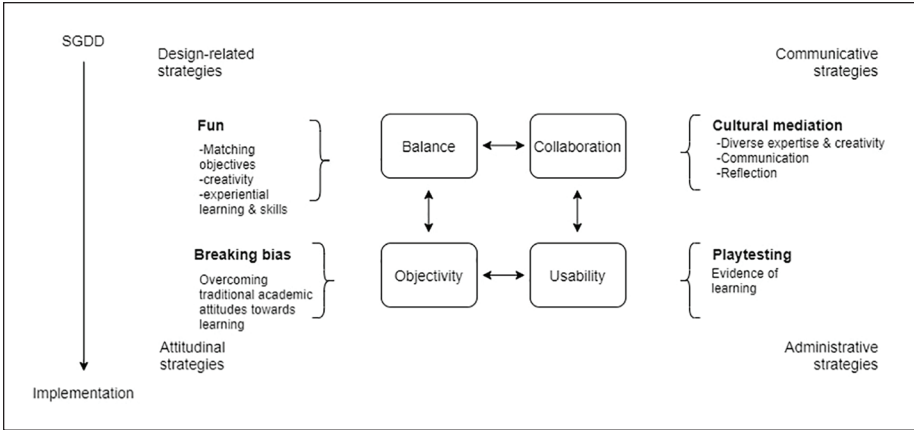


Figure 2. Effective strategies for SGDD.

concepts into learning (like win/lose, success factors, processes etc.) for more immersive experiences". If the game does not help learners to develop a particular set of skills or knowledge, it will not be used. For this, creativity and a focus on experiential learning and skills acquisition are needed.

Regarding the perceived need to create fun games, references were made to SG that integrate learning with entertainment by catering for a diversity of entertaining activities in order to appeal to wider audiences. This also allows for easier player management. Accordingly, the fun factor can be achieved by games that aim to *"tackle all age groups"*, *"have[ing] multiple back-up activity options for different learners and their abilities"* and *"make[ing] them engaging for the learner"*. This is achieved through team collaboration, identified as another essential element of a SGDD strategy with reference to the multidisciplinary nature of SGDD teams. Participants expressed the view that an effective team is a *"good and willing team"* with creativity and diverse expertise. According to one participant's experience:

There was a challenge of bringing together different professions to create the virtual gaming simulation. We came with different ideas and perspectives. But, it was also our strength, without all the different team members we could not have created such an effective learning tool.

Accordingly, a SGDD team needs to be composed of a pool of trained professionals who need to convey their design approach to developers. Key issues and decisions need to be clearly communicated within the team. The need for communication is thus characterized by a constructive review throughout the SGDD process, and presupposed by the team's need to *"believe in the task"* - with special reference to *"the need to fully convey design to developers"*. Subsequently, the effectiveness of communication is

perceived as requiring cultural mediation, or the need to acculturate team members into other team members' working cultures, expertise, and practices:

The main issue is intercultural communication [between team members] who may or may not understand where the others are coming from. We appointed a cultural mediator who was our go to person for explaining and helping solve these sorts of communication issues and to give us insights into finding ways to collaborate. This person is like a Guru/Psy/Tech and Confessor all in one.

Additionally, communication within the team can be effective *“if all parties on the development team know and buy into the model at the outset”*, emphasizing the importance of a clear vision at the start of the design process. What is more, as educators, participants believed that the effectiveness of a SGDD strategy is underpinned by the need for continuous reflection, as *“without reflection, you don't have learning”*.

For successful implementation at the post-production stage of SGDD, it is important to highlight the *“usability and sustainability”* of a SG and provide stakeholders with evidence that the game is current, relevant to target audiences, and effective for learning. Hence, playtesting is an important strategy not only for refining a SG, but also *“selling”* the idea to administration, faculty, and students. However, none of the participants went into detail as to how this is achieved at their institutions.

Lastly, if there is going to be a mainstream implementation of SG by higher education institutions, objectivity needs to be created in the academic community. For this, participants in this study agreed that the educator has to break barriers related to bias with regard to the process of learning through fun. However, similar to usability, no educator outlined a specific strategy that would help them create objectivity about the usefulness of SG as teaching and learning tools.

Discussion

In this article we explored educators' perceived challenges in SGDD and the strategies deployed to overcome those. While past research points to challenges at the institutional, social, and personal level, we also identified challenges related to the production level and reclassified these into administrative, communicative, attitudinal, and design-related challenges. To overcome these, educators tend to deploy effective strategies related to the SGDD stages of concept development, pre-production, and production – namely strategies to overcome design-related and communicative challenges. Meanwhile, although challenges are acknowledged and some strategies are deployed at the post-production stage, with reference to implementation, it is not perfectly clear what those strategies are. During the first four SGDD stages, educators use different tactics to achieve a balance between learning and fun, as well as team collaboration. However, as sole actors of implementation, educators need to create a culture of objectivity towards the use of serious games in the academia. Their current approach uses usability studies to convince stakeholders and target audiences about SG learning effectiveness.

We suggest that future SGDD processes and methodologies need be addressed with clear allocation of roles to members of multidisciplinary teams. The stages of concept development, pre-production and production can be improved by adopting approaches and tools that enhance communication between multidisciplinary team members. These tools should emphasize on generating diverse ideas, such as our in-house methodology (Digital Education Strategies, 2018). The stage of post-production can be improved, firstly, by developing effective administrative and attitudinal strategies; and secondly, by easing the burden placed on educators involved in SGDD. This can be achieved through the update of game design curricula and training to include a focus on communication skills and idea generation. Also, the use of SG as teaching tools can be made more attractive to faculty if a stronger focus is placed on implementation, and includes clear directives regarding the usability of SG, as well as guidelines for achieving objectivity towards SG within the academic institutions. Strategies for the attainment of these should focus on the development of trust between educators as SG implementers, stakeholders, and target audiences. Factors instrumental in the development of trust towards a game product include transparency about the strengths and shortcomings of a SG, the disclosure of how game objectives and learning objectives are linked to a given curriculum, and how a particular SG promotes knowledge and skills acquisition.

Limitations and Future Research

The limitation of this study is that secondary analysis could only provide us with qualitative information, as the primary data were collected for the purposes of developing a guide to SG for multidisciplinary teams. Hence, our findings could be confirmed by a follow-up quantitative study.

Further research into the dynamics of socialization and acculturation between members of multidisciplinary teams is needed, and may give rise to new SG design working practices that overcome communicative challenges. Action research exploring attitudinal barriers towards SG use as teaching and learning tools may provide additional insights in how to create objectivity towards SG adoption in the wider academic community.

Conclusion

This study investigated educators' perceptions of challenges related to the SGDD process and strategies deployed to overcome these. We classified challenges experienced as administrative, communicative, attitudinal, and design-related challenges, and highlighted that educators deploy effective strategies at the concept development, pre-production and production stages of SGDD. However, at the post-production stage, a clear pattern in challenges and mitigating strategies could not be observed. We suggest that future SGDD processes need to address these challenges, and strategies adopted should focus on the development of trust between educators in their role as SG implementers stakeholders, and target audiences.

At a time where SG are increasingly popular with many games freely available as OER, in parallel with increasing preference of online learning, especially in crisis response situations such as the current COVID-19 lockdown, SGDD approaches need to be revisited and revised to incorporate SG usability and stakeholder/audience objectivity. At the same time, post-secondary education institutions need to support this effort through the provision of resources, professional development and incentives for faculty, and instructors to use SG in their teaching.

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References

- Aleem, S., Capretz, L. F., & Ahmed, F. (2016). Game development software engineering process life cycle: A systematic review. *Journal of Software Engineering Research and Development*, 4(6), 1–30. <https://doi.org/10.1186/s40411-016-0032-7>
- Annetta, L. A., Minogue, J., Holmes, S. Y., & Cheng, M. T. (2009). Investigating the impact of video games on high school students' engagement and learning about genetics. *Computers & Education*, 53(1), 74–85. <https://doi.org/10.1016/j.compedu.2008.12.020>
- Arnab, S., & Clarke, S. (2017). Towards a trans-disciplinary methodology for a game-based intervention development process. *British Journal of Educational Technology*, 48(2), 279–312. <https://doi.org/10.1111/bjet.12377>
- Arnab, S., Lim, T., Carvalho, M. B., Bellotti, F., De Freitas, S., Louchart, S., Suttie, N., Berta, R., & De Gloria, A. (2015). Mapping learning and game mechanics for serious games analysis. *British Journal of Educational Technology*, 46(2), 391–411. <https://doi.org/10.1111/bjet.12113>
- Bergeron, B. P. (2008). Learning & retention in adaptive serious games. *Studies in Health Technology and Informatics*, 132, 26–30.
- Bleumers, L., Mariën, I., Van Looy, J., Stewart, J., Schurmans, D., & All, A. (2013). Best practices for deploying digital games for personal empowerment and social inclusion. In P. Escudeiro & C. DeCarvalho (Eds.), *7th European Conference on Games Based Learning (ECGBL)* (Vol. 1, pp. 53–61). Porto, Portugal: ACAD.

- Brandão, J., Ferreira, T., & Carvalho, V. (2012). An overview on the use of serious games in the military industry and health. In M. M. Cruz-Cunha (Ed.), *Handbook of research on serious games as educational, business and research tools* (pp. 182–201). IGI Global. <https://doi.org/10.4018/978-1-4666-0149-9.ch009>
- Chan, K., & Zary, N. (2019). *Knowledge, use and barriers to Serious Games in undergraduate medical education*. OSF Preprints. <https://doi.org/10.31219/OSF.IO/2F7RW>
- Clarke, J., Nelson, K., & Stoodley, I. (2013). *The place of higher education institutions in assessing student engagement, success and retention* [Conference session]. *Proceedings of the Higher Education Research and Development Society of Australasia Conference*. <https://eprints.qut.edu.au/60024/16/60024.pdf>
- de Freitas, S. (2006). Learning in immersive worlds: A review of game-based learning. *JISC E-Learning Programme*, 3, 73. http://www.jisc.ac.uk/uploaded_documents/Summary_report
- De Gloria, A., Bellotti, F., & Berta, R. (2014). Serious Games for education and training. *International Journal of Serious Games*, 1(1). <https://doi.org/10.17083/ijsg.v1i1.11>
- De Grove, F., Bourgonjon, J., & Van Looy, J. (2012). Digital games in education? A contextual approach to teachers' adoption intention of digital games for learning purposes. *Computers in Human Behavior*, 28(6), 2023–2033.
- Digital Education Strategies. (2018). *The art of serious game design*. The Chang School of Continuing Education, Ryerson University. <https://pressbooks.library.ryerson.ca/guide/front-matter/introduction/>
- Duke, R. D. (1980). A paradigm for game design. *Simulation & Games*, 11(3), 364–377. <https://doi.org/10.1177/104687818001100308>
- Easterly, J. L. (1978). Simulation game design—a philosophic dilemma. *Simulation & Games*, 9(1), 23–28. <https://doi.org/10.1177/003755007891002>
- Groff, J., Clarke-Midura, J., Owen, V., & Rosenheck, L. (2015). *Better learning in games: A balanced design lens for a new generation of learning games*. Learning Games Network, MIT Education Arcade. https://digitalcommons.usu.edu/itls_facpub/540/
- Guillén-Nieto, V., & Aleson-Carbonell, M. (2012). Serious games and learning effectiveness: The case of *It's a Deal!* *Computers & Education*, 58, 435–448. <https://doi.org/10.1016/j.compedu.2011.07.015>
- Hall, J. J. (2014). Existing and emerging business simulation-game design movements. In *Developments in business simulation and experiential learning: Proceedings of the Annual ABSEL Conference (Vol. 36)*. <https://absel-ojs-ttu.tdl.org/absel/index.php/absel/article/view/350/316>
- Heaton, J. (2004). *Reworking qualitative data*. Sage. <https://doi.org/10.4135/9781849209878>
- Hirumi, A., & Stapleton, C. (2008). Applying pedagogy during game development to enhance game-based learning. In C. Miller (Ed.), *Games: Purpose and potential in education* (pp. 127–162). Springer. https://doi.org/10.1007/978-0-387-09775-6_6
- Huizenga, J. C., ten Dam, G. T. M., Voogt, J. M., & Admiraal, W. F. (2017). Teacher perceptions of the value of game-based learning in secondary education. *Computers & Education*, 110, 105–115. <https://doi.org/10.1016/j.compedu.2017.03.008>
- Iten, N., & Petko, D. (2016). Learning with serious games: Is fun playing the game a predictor of learning success? *British Journal of Educational Technology*, 47(1), 151–163. <https://doi.org/10.1111/bjet.12226>
- Kapralos, B., Hogan, M., Pribetic, A. I., & Dubrowski, A. (2011). Virtual simulations and serious games in a laptop-based university: Gauging faculty and student perceptions. *Interactive Technology and Smart Education*, 8(2), 106–120. <https://doi.org/10.1108/17415651111141821>

- Kebritchi, M. (2010). Factors affecting teachers' adoption of educational computer games: A case study. *British Journal of Educational Technology*, 41(2), 256–270. <https://doi.org/10.1111/j.1467-8535.2008.00921.x>
- Kenny, R., & Gunter, G. (2011). Factors affecting adoption of video games in the classroom. *Journal of Interactive Learning Research*, 22(2), 259–276.
- Kirkley, J., Kirkley, S., & Heneghan, J. (2007). Building bridges between serious game design and instructional design. In *The design and use of simulation computer games in education* (pp. 61–83). Brill | Sense. https://doi.org/10.1163/9789087903121_006
- Klabbers, J. (2006). A framework for artifact assessment and theory testing. *Simulation & Gaming*, 37(2), 155–173. <https://doi.org/10.1177/1046878106287943>
- Korhonen, T., Halonen, R., Ravelin, T., Kempainen, J., & Koskela, K. (2017, September). A multidisciplinary approach to serious game development in the health sector [Conference session]. *The 11th Mediterranean Conference on Information Systems (MCIS)*, Genoa, Italy. <https://aisel.aisnet.org/cgi/viewcontent.cgi?article=1016&context=mcis2017>
- Lean, J., Moizer, J., Towler, M., & Abbey, C. (2006). Simulations and games. *Active Learning in Higher Education*, 7(3), 227–242. <https://doi.org/10.1177/1469787406069056>
- Mayer, I. (2012). Towards a comprehensive methodology for the research and evaluation of serious games. *Procedia Computer Science*, 15, 233–247. <https://doi.org/10.1016/J.PROCS.2012.10.075>
- Moizer, J., Lean, J., Towler, M., & Abbey, C. (2009). Simulations and games: Overcoming the barriers to their use in higher education. *Active Learning in Higher Education*, 10(3), 207–224. <https://doi.org/10.1177/1469787409343188>
- Muntean, C. C. I. (2011). Raising engagement in e-learning through gamification. *The 6th International Conference on Virtual Learning ICVL 2011*, 1, 323–329. http://icvl.eu/2011/disc/icvl/documente/pdf/met/ICVL_ModelsAndMethodologies_paper42.pdf
- Noraddin, E. M., Tse Kian, N., & Abdulrahman, N. B. (2013). Academics' attitudes toward using digital games for learning & teaching in Malaysia. *Malaysian Online Journal of Educational Technology*, 2(4). <http://www.mojet.net/frontend/articles/pdf/v02i04/v02i04-01.pdf>
- Peppers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>
- Pitts, K., & Ashwin, A. (2005). Online games and simulations as aids to learning ethos, challenges and evaluation. *Interact*, 31, 12–13.
- Proctor, M. D., & Marks, Y. (2013). A survey of exemplar teachers' perceptions, use, and access of computer-based games and technology for classroom instruction. *Computers & Education*, 62, 171–180. <https://doi.org/10.1016/j.compedu.2012.10.022>
- Rankin, Y. A., McNeal, M., Shute, M. W., & Gooch, B. (2008). User centered game design: Evaluating massive multiplayer online role playing games for second language acquisition. In *Proceedings of Sandbox 2008: An ACM SIGGRAPH Videogame Symposium, Sandbox '08* (pp. 43–50). <https://doi.org/10.1145/1401843.1401851>
- Rooney, P. (2012). A theoretical framework for serious game design: Exploring pedagogy, play and fidelity and their implications for the design process. *International Journal of Game-Based Learning*, 2, 41–60. <https://arrow.dit.ie/lcart/28>
- Ruggiero, D., & Watson, W. R. (2014). Engagement through praxis in educational game design: Common threads. *Simulation & Gaming*, 45, 471–490. <https://doi.org/10.1177/1046878114553570>

- Smith, K., Shull, J., Shen, Y., Dean, A., & Heaney, P. (2018). A framework for designing smarter serious games. In *Smart Innovation, Systems and Technologies*. https://doi.org/10.1007/978-3-319-59454-5_9
- Staalduinen, J. P., & de Freitas, S. (2011). A game-based learning framework: Linking game design and learning outcomes. In M. S. Khyne (Ed.), *Learning to play: Exploring the future of education with video games*. Peter Lang. [https://books.google.ca/books?hl=en&lr=&id=FIC1w66WFIAC&oi=fnd&pg=PA29&dq=Van+Staalduinen,+J.+P.,+%26+de+Freitas,+S.+\(2011\).+A+game-based+learning+framework:+Linking+game+design+and+learning.+Learning+to+play:+exploring+the+future+of+education+with+video+Stewart,+J.,+Lizzy,+I.,+All,+A.,+Mariën,+I.,+Schurmans,+D.,+Looy,+V.,+Jacobs,+A.,+Willaert,+K.,+&+De+Grove,+F.+\(2013\).+Digital+games+for+empowerment+and+inclusion+of+groups+at+risk+of+social+and+economic+exclusion:+Evidence+and+opportunity+for+policy.+European+Commission+Joint+Research+Centre.+https://doi.org/10.2791/88148](https://books.google.ca/books?hl=en&lr=&id=FIC1w66WFIAC&oi=fnd&pg=PA29&dq=Van+Staalduinen,+J.+P.,+%26+de+Freitas,+S.+(2011).+A+game-based+learning+framework:+Linking+game+design+and+learning.+Learning+to+play:+exploring+the+future+of+education+with+video+Stewart,+J.,+Lizzy,+I.,+All,+A.,+Mariën,+I.,+Schurmans,+D.,+Looy,+V.,+Jacobs,+A.,+Willaert,+K.,+&+De+Grove,+F.+(2013).+Digital+games+for+empowerment+and+inclusion+of+groups+at+risk+of+social+and+economic+exclusion:+Evidence+and+opportunity+for+policy.+European+Commission+Joint+Research+Centre.+https://doi.org/10.2791/88148)
- Stewart, J., Lizzy, I., All, A., Mariën, I., Schurmans, D., Looy, V., Jacobs, A., Willaert, K., & De Grove, F. (2013). *Digital games for empowerment and inclusion of groups at risk of social and economic exclusion: Evidence and opportunity for policy*. European Commission Joint Research Centre. <https://doi.org/10.2791/88148>
- Swier, R., & Peterson, M. (2018). 3D digital games, virtual worlds, and language learning in higher education: Continuing challenges in Japan. *JALT CALL Journal*, 14(3), 225–238. <https://doi.org/https://doi.org/10.29140/jaltcall.v14n3.232>
- To, A., Fath, E., Zhang, E., Ali, S., Kildunne, C., Fan, A., Hammer, J., & Kaufman, G. (2016, October). *Tandem transformational game design: A game design process case study* [Conference session]. *Proceedings of the International Academic Conference on Meaningful Play*. <https://static1.squarespace.com/static/5a4aae77e5dd5b62b3ebfbc7/t/5ac1003a2b6a28a9261a0b5d/1522597951577/TandemDesignPaper.pdf>
- Ucus, S. (2015). Elementary school teachers' views on game-based learning as a teaching method. *Procedia—Social and Behavioral Sciences*, 186, 401–409. <https://doi.org/10.1016/j.sbspro.2015.04.216>
- Wilkinson, P. (2016). Brief history of serious games. In *Lecture notes in computer science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics) (Vol. 9970, pp. 17–41)*. https://doi.org/10.1007/978-3-319-46152-6_2
- Wronowski, M., Urick, A., Wilson, A. S. P., Thompson, W., Thomas, D., Wilson, S., Elizondo, F. J., & Ralston, R. (2020). Effect of a serious educational game on academic and affective outcomes for statistics instruction. *Journal of Educational Computing Research*, 57, 2053–2084. <https://doi.org/10.1177/0735633118824693>
- Zarraonandia, T., Diaz, P., Aedo, I., & Ruiz, M. R. (2015). Designing educational games through a conceptual model based on rules and scenarios. *Multimedia Tools and Applications*, 74(13), 4535–4559. <https://doi.org/10.1007/s11042-013-1821-1>
- Zook, A., & Riedl, M. O. (2013, May). *Game conceptualization and development processes in the Global Game Jam* [Conference session]. Workshop Proceedings of the 8th International Conference on the Foundations of Digital Games. http://www.fdg2013.org/program/workshops/papers/GGJ2013/ggj13_submission_4.pdf

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